BlueArkive.

The Sovereign AI Edge Node

The Surveillance Deficit

Modern AI meeting transcribers rely exclusively on remote compute. This structural flaw renders them unusable in heavily regulated enterprise environments.

The Status Quo

A third-party bot joins the conference call, streams unencrypted proprietary audio to a cloud LLM, and violates strict GDPR/HIPAA compliance.

The Sovereign Edge

All audio is ingested natively across OS-level hooks and transcribed via hardware acceleration locally on the user's CPU, maintaining absolute data privacy.

Engineering The Native Edge

BlueArkive functions as a hyper-optimized, zero-network desktop application. By relying on native C++ buffers, we avoid the lethal RAM spikes of typical Electron apps.

Zero-Copy Dual Audio

We bypass standard browser APIs by hooking simultaneously into system output and mic input using `SharedArrayBuffer` memory, driving memory-leak risk to zero during 3-hour sessions.

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Voice Activity Detection

To preserve CPU, an independent `Silero VAD` worker thread drops silent frames instantly, cutting overall inference workload by 40% before audio touches the transcription engine.

Core Architecture Flow

Our proprietary stack executes highly concurrent LLM processing entirely offline.

OS Audio
SharedArrayBuffer

▶

Silero VAD
Speech Filter

▶

Whisper C++
51.8x Real-Time

▶

SQLite WAL
Memory State

▶

Qwen 2.5 3B
Local Expansion

Audio is piped natively through the C++ transcription engine directly into local SQLite memory.

Model Ensemble Matrix

Deploying massive LLMs on commercial hardware requires extreme restraint. We benchmarked 40+ local foundation models to synthesize the ultimate fallback ensemble.

The Phase 0 Hardware Loadout (16GB RAM Class):

- **Speech-to-Text:** Whisper Large V3 `Turbo` (C++)
  - *Benchmark*: Processes a 30-second audio stream in exactly **0.58 seconds** (51.8x real-time).
- **Natural Language Processor:** Qwen 2.5 3B (MLX optimized)
  - *Benchmark*: Operates highly structured data expansion inside a 32k context window utilizing merely **2.2GB** of RAM footprint.

Offline Entity Architecture

Intelligence requires structured data. As a meeting progresses, the local engine identifies critical metadata in real-time.

It automatically utilizes regex and local parsing arrays to extract:

- **Financial Thresholds & Budgets**
- **Assigned Tasks & Deadlines**
- **Key Personnel & Organizations**

These arrays are parsed into interactive "Smart Chips" inside the UI, letting the user instantly view every decision tied to a specific deadline across historical offline meetings.

The Encrypted RAG Paradox

How do you perform offline semantic search across multiple devices while keeping data completely blind to the server?

1. **Local Vectors First:** The app deploys `all-MiniLM-L6-v2` locally to cast 384-dimensional mathematical arrays of your meetings onto your physical drive.
2. **Military Grade Locking:** The raw text notes are subsequently locked in **AES-256-GCM** ciphertext.
3. **The Result:** Your client search operates instantly and offline via relative cosine similarity, while the actual encrypted data can sync via the cloud securely without any server being able to read it.

7. The Technology Stack

We didn't just connect APIs. We engineered a robust, 65,000+ line local intelligence fabric.

- Electron
- React
- TypeScript
- SharedArrayBuffers (C++)
- Hardware VAD Hooks
- SQLite (WAL Mode)
- AES-256-GCM
- Whisper Turbo (GGUF)
- Qwen 2.5 (3B)

> Privacy is not a premium feature.
> It is the default architecture.